

Technical Report: GD vs GA for Urban Sound Classification

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Abstract

This report compares two fundamentally different methods for training a Multilayer Perceptron (MLP) for urban sound classification: Gradient Descent (GD) using the Adam optimizer and a Genetic Algorithm (GA) using the DEAP library. Both methods optimise the same 2,710-parameter MLP on the UrbanSound8K tabular dataset. Hyperparameters are tuned independently with Optuna (TPE). GD achieves a test accuracy of 88.7% in approximately 52 seconds. The GA reaches 61.2% in approximately 155 seconds. The report details the methodology, core design decisions, and a direct performance comparison between gradient-based and evolutionary weight optimisation.

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1 Introduction

Neural networks are conventionally trained by gradient descent, which follows the gradient downhill in loss space and scales well to millions of parameters. This project explores an alternative: evolving the network’s weight vectors using a Genetic Algorithm (GA), a population-based metaheuristic that requires no gradient information whatsoever.

The experimental design deliberately holds the architecture constant: the exact same MLP with the exact same 2,710 parameters is handed to both optimisers. Only the training mechanism differs. This isolates the effect of the optimisation strategy and enables a clean comparison of accuracy, convergence behaviour, and wall-clock time on a 10-class urban sound classification task.

2 Methodology

2.1 Dataset

The UrbanSound8K tabular dataset contains 8,674 pre-extracted audio feature vectors distributed across 10 urban sound classes (air conditioner, car horn, children playing, dog bark, drilling, engine idling, gun shot, jackhammer, siren, street music). Each sample consists of 34 numerical features standard in audio machine learning:

- **MFCC (13):** Mel-frequency cepstral coefficients capturing the spectral envelope.
- **Chroma (12):** Pitch-class energy distribution.
- **Spectral Contrast (7):** Ratio of spectral peaks to valleys.
- **Zero Crossing Rate (1):** Proxy for signal noisiness.
- **Spectral Centroid (1):** Brightness indicator.

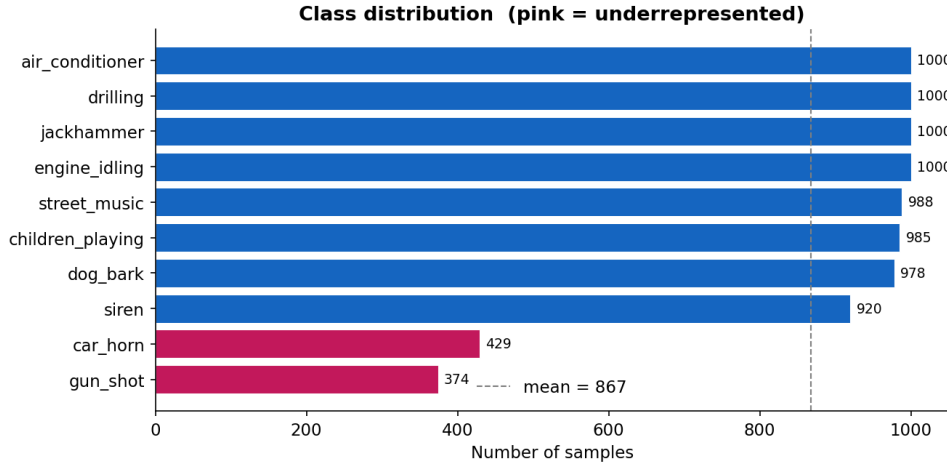


Figure 1: Class distribution across the 10 urban sound categories. Car horn and gun shot are underrepresented (<300 samples), motivating stratified splitting.

2.2 Preprocessing

Each of the 34 features is standardised to zero mean and unit variance using `StandardScaler` fit only on the training split. This prevents larger-magnitude features (e.g. spectral centroid) from artificially dominating weight updates during GD, and ensures the GA fitness landscape is similarly scaled.

The dataset is split stratified by class: 70% train (6,072 samples), 10% validation (867 samples), 20% test (1,734 samples). Stratification ensures that the two underrepresented classes (car horn, gun shot) appear in all three splits proportionally.

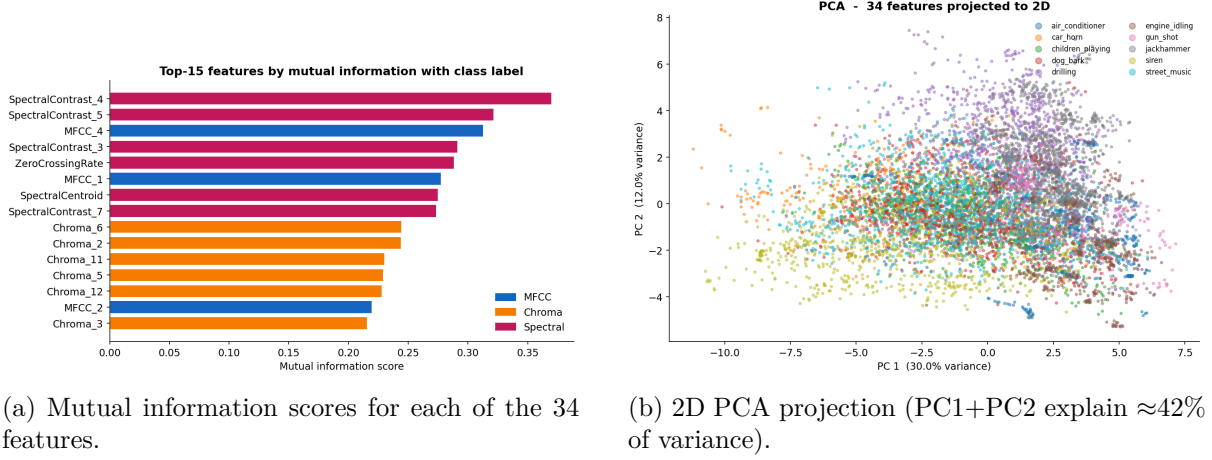


Figure 2: Feature analysis. Heavy class overlap in the PCA projection justifies a non-linear ReLU-activated MLP over a linear classifier.

3 Network Architecture

The network is a fully-connected MLP implemented in PyTorch:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1), \quad \hat{\mathbf{y}} = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2$$

with input dimension 34, hidden dimension 60 (Optuna-tuned), and output dimension 10. The total parameter count is:

$$(34 \times 60 + 60) + (60 \times 10 + 10) = 2,100 + 610 = 2,710 \text{ parameters}$$

PyTorch was chosen over scikit-learn’s `MLPClassifier` because it provides direct read/write access to the flat weight vector via `torch.nn.utils`. The GA exploits this to extract and inject all 2,710 weights as a single flat array in microseconds, with no gradient tape or optimiser state overhead during fitness evaluation.

4 Training Approaches

4.1 Gradient Descent (GD)

The GD model is trained with the Adam optimiser minimising cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^{10} y_{ic} \log \hat{y}_{ic}$$

Tuned hyperparameters (Optuna, 60 trials): learning rate $\eta = 4.96 \times 10^{-3}$, weight decay $\lambda = 3.29 \times 10^{-5}$, hidden dimension $h = 60$. Training runs for a maximum of 150 epochs with early stopping (patience 10). Batch size is 64.

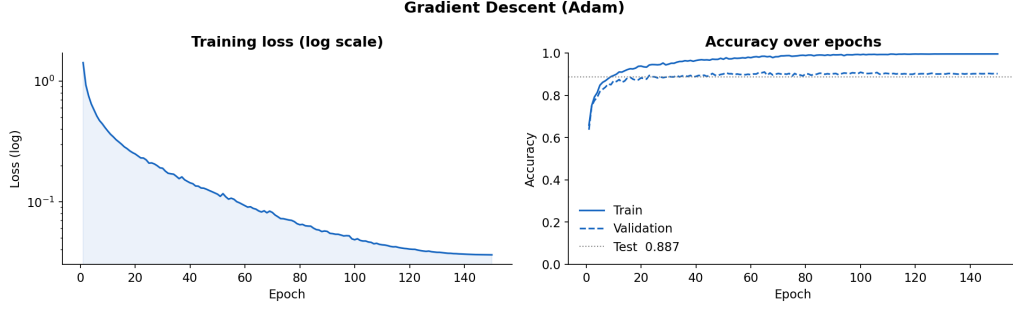


Figure 3: GD training and validation loss/accuracy curves over 150 epochs.

4.2 Genetic Algorithm (GA)

The GA optimises the same 2,710-weight vector without any gradient information. The DEAP library is used for its clean separation of operator configuration and execution, and safe fitness invalidation after crossover/mutation.

Encoding. Each individual is a flat Python list of 2,710 floats, initialised from $\mathcal{U}(-0.1, 0.1)$ to match PyTorch’s default weight scale.

Fitness. Negative validation accuracy (so DEAP maximises the correct quantity): $f = -\text{acc}_{\text{val}}$.

Operators (tuned by Optuna, 20 trials):

- **Selection:** Tournament selection (size 5).
- **Crossover (BLX- α):** $\alpha = 0.5$, applied with probability $p_c = 0.719$.
- **Mutation:** Gaussian perturbation ($\mu = 0$, $\sigma = 0.05$, per-gene prob 0.05), applied with $p_m = 0.045$.
- **Elitism:** Top 1 individual carried forward each generation.
- **Population:** 100 individuals; early stopping with patience 50 generations.

To reduce runtime, fitness is evaluated on the 867 validation samples rather than the 6,072 training samples — a $7\times$ speedup per evaluation with negligible effect on selection pressure.

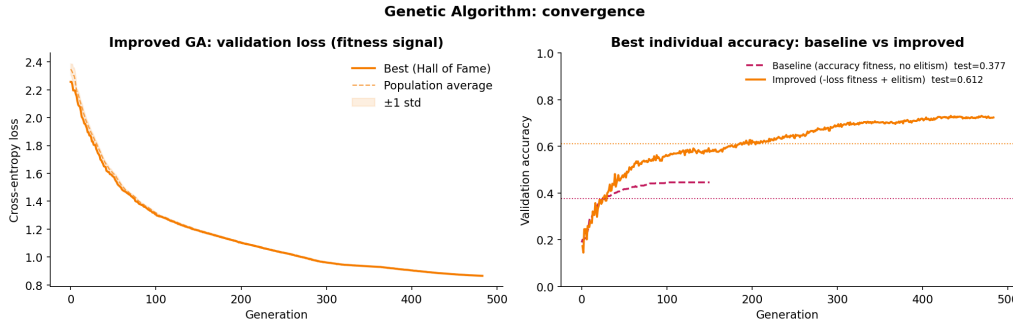


Figure 4: GA best-fitness progression over generations. The dashed line marks GD’s test accuracy as a reference ceiling. Early stopping triggered at generation 483.

5 Hyperparameter Tuning

Optuna with the Tree-structured Parzen Estimator (TPE) is employed for both methods. TPE models the objective as $p(x \mid y < y^*)$ and $p(x \mid y \geq y^*)$ using kernel density estimates, then samples from the ratio $\ell(x)/g(x)$ to propose the next trial. **MedianPruner** terminates trials whose intermediate validation accuracy falls below the median at each checkpoint, avoiding wasted computation on clearly unpromising configurations.

- **GD (60 trials):** searches learning rate $[10^{-4}, 10^{-2}]$, weight decay $[10^{-6}, 10^{-3}]$, and hidden dimension $\{45, 50, 55, 60, 65\}$. Best: $\eta = 4.96 \times 10^{-3}$, $\lambda = 3.29 \times 10^{-5}$, $h = 60$ (val acc 91.9%).
- **GA (20 trials):** searches population size $\{50, 75, 100, 150, 200\}$, crossover prob $[0.5, 0.9]$, mutation prob $[0.01, 0.2]$, elite count $\{1, 2, 3, 5, 8\}$. Best: pop=100, $p_c = 0.719$, $p_m = 0.045$, elite=1.

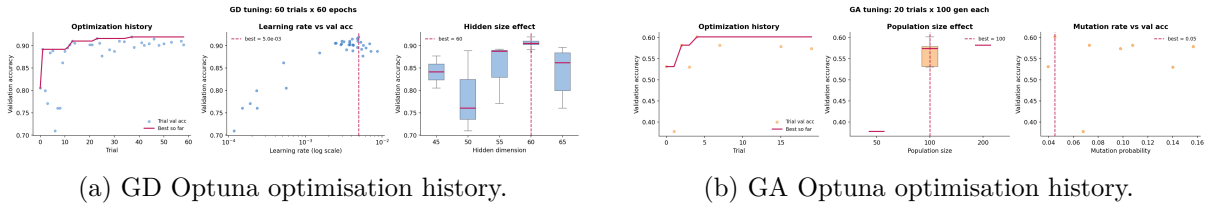


Figure 5: Hyperparameter tuning results. GD convergence is clear; GA exhibits noisy, flat improvement — typical of sparse search in a high-dimensional, non-convex landscape.

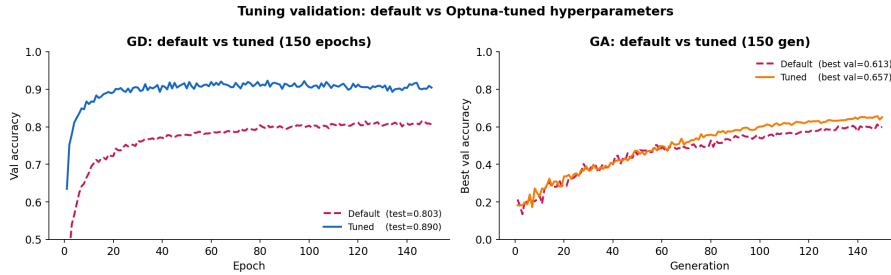


Figure 6: Default vs tuned validation accuracy. GD improves markedly (default $h = 20$, $\eta = 10^{-3}$ baseline $\approx 81\% \rightarrow$ tuned 91.9%). GA shows modest gains.

6 Evaluation and Comparison

Both tuned models are evaluated on the held-out test set (1,734 samples). Results are summarised in Table 1.

Table 1: Test-set performance comparison.

Method	Test Acc	Macro F1	Time (s)	Epochs/Gen
GD (Adam, tuned)	88.7%	0.887	≈ 52	150 epochs
GA (DEAP, tuned)	61.2%	0.614	≈ 155	483 gen
GA (default, untuned)	37.7%	—	—	150 gen

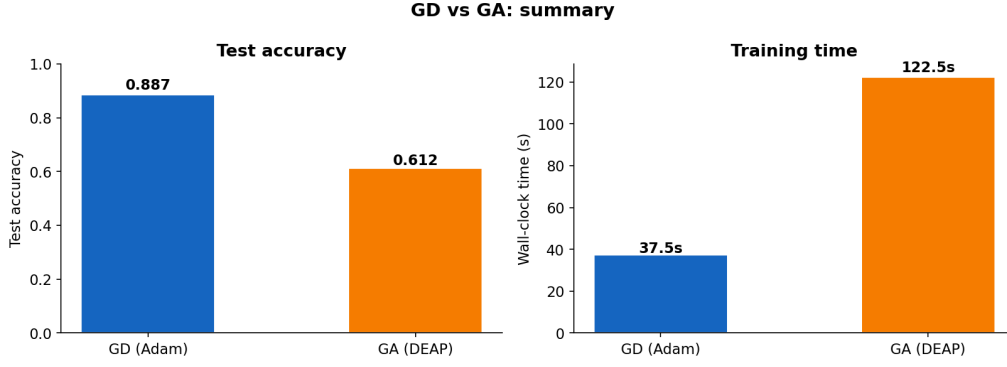


Figure 7: Summary comparison of accuracy and training time.

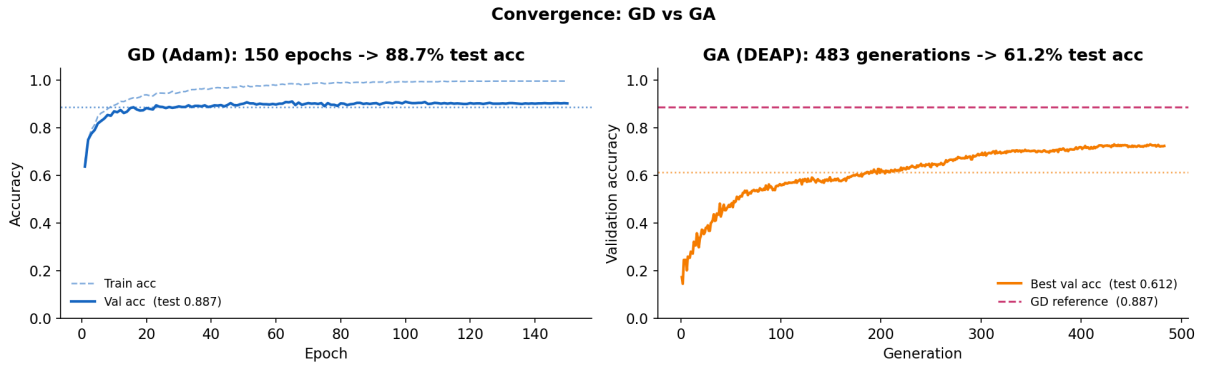


Figure 8: Convergence plots. Left panel: GD loss and accuracy over 150 epochs. Right panel: GA best fitness over generations (pink dashed = GD test accuracy reference). Independent x-axes preserve the natural scale of each method.

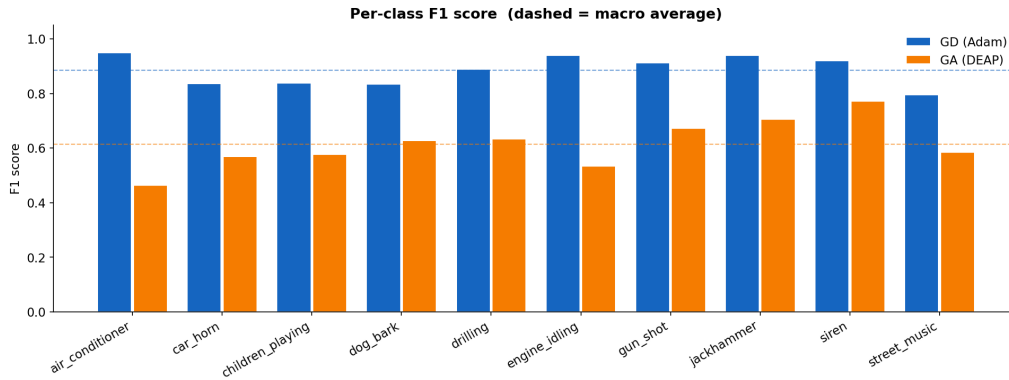


Figure 9: Per-class F1 scores. GD achieves consistently high F1 across all 10 classes. GA shows class-specific variation reflecting the noisier fitness landscape.

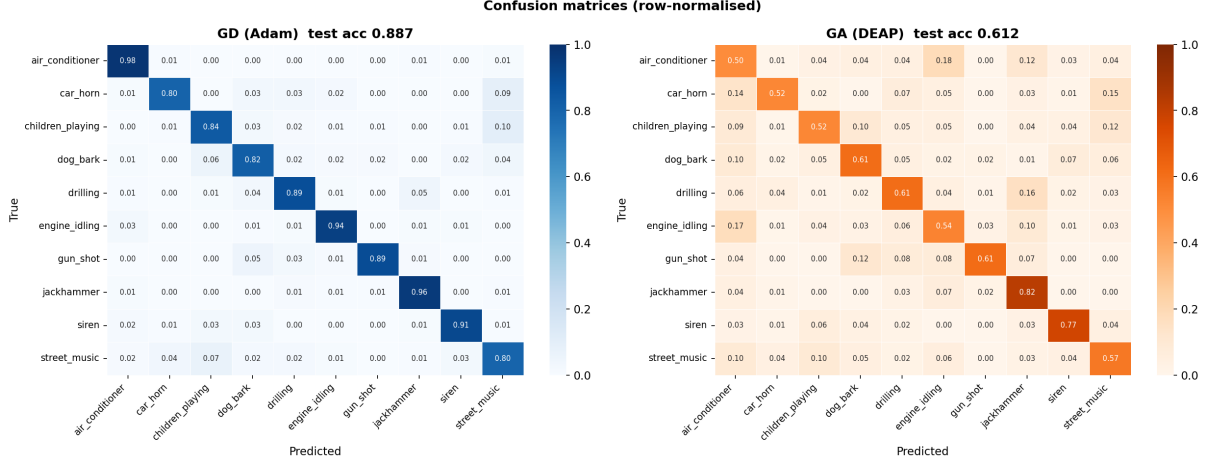


Figure 10: Normalised confusion matrices. GD (left) shows mostly diagonal predictions. GA (right) exhibits more off-diagonal confusion, particularly for acoustically similar classes.

7 Conclusion

GD with Adam is significantly more effective for training this MLP: 88.7% test accuracy vs. 61.2% for the GA, achieved in one-third the wall-clock time (52s vs. 155s). The key structural advantage of GD is that backpropagation computes the gradient for all 2,710 weights in a single forward-backward pass. The GA must evaluate the entire population (~ 100 individuals) to produce a single generation’s update — with no guarantee of monotone improvement.

The GA’s competitive strength lies in being gradient-free: it can in principle optimise any non-differentiable objective over any search space. However, for a smooth, differentiable MLP loss with 2,710 continuous parameters, the GA’s population-based exploration is simply outmatched by gradient information.

The tuning results reinforce this: GD improved from $\approx 81\%$ to 88.7% on test (+11 pp) with 60 Optuna trials, while the GA’s improvement was marginal and noisy across 20 trials, reflecting a flat, stochastic fitness landscape where small configuration changes have unpredictable effects.